

MOHAMED MAGDY

AI Engineer

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Professional Summary

AI Engineer and Data Scientist specializing in Generative AI, Computer Vision, and Industry 4.0 systems. Experienced in engineering advanced standard and Graph RAG pipelines, designing LLM-powered workflows, and deploying highly optimized hybrid search solutions. Proven ability to translate complex AI architectures—from Digital Twins to enterprise-grade Document Intelligence—into measurable, production-ready business value.

Technical Skills

Generative AI & Search: OpenAI, LLMs (GPT, LLaMA), AI Agents, Prompt Engineering (Few-Shot), Standard & Graph RAG, Vector Databases, Hybrid Search (HNSW, BM25), Advanced Chunking, LangChain, LangSmith
Cloud & Infrastructure: Microsoft Azure, Azure OpenAI, Azure AI Services, Docker, Git, CI/CD
Backend & Automation: FastAPI, Redis Caching, n8n
Machine Learning & CV: CNNs, Transformers, YOLO, SAM, Scikit-Learn, TensorFlow, PyTorch
Data & BI: Python, SQL, Pandas, NumPy, Power BI, Tableau, Excel
Engineering: Digital Twins, Predictive Maintenance

Experience

Global Brands Group (GBG) Feb 2026 – Present
AI Engineer Intern (Client Solutions & GenAI) Cairo, Egypt

- Selected for a 6-month client-facing extension after graduating as a Top Achiever (Score: 950/950) and earning a Certificate of Achievement with Highest Commendation.
- Collaborating directly with core engineering teams to develop and deploy scalable enterprise AI solutions for real-world clients.
- Architected an end-to-end, production-ready “Airport Lost and Found System,” integrating LLMs, Graph RAG, FastAPI, and Redis caching, deployed via Microsoft Azure.
- Designed high-accuracy retrieval pipelines utilizing advanced chunking methods, Vector Databases, and Hybrid Search algorithms (HNSW & BM25) alongside LangChain and LangSmith.

Siemens Industry 4.0 Innovation Center (ITIDA) Jan 2026 – March 2026
AI & Digital Twin Engineer Intern New Administrative Capital, Egypt

- Developing Digital Twin simulations to model, visualize, and optimize manufacturing workflows.
- Implementing AI-based monitoring systems for smart factory environments to reduce downtime.
- Designing predictive maintenance algorithms using real-time sensor data.

National Telecommunication Institute (NTI) Oct 2025 – Jan 2026
Digital Twin Engineering Trainee Cairo, Egypt

- Built end-to-end predictive maintenance models using ML to forecast equipment failures.
- Integrated AI models into real-time dashboards, providing actionable insights for engineering teams.
- Optimized industrial workflow simulations, reducing theoretical process latency.

ContiLab, Chung-Ang University (CAU) Mar 2025
(Fully Funded Scholarship) Seoul, South Korea

- Selected for a competitive, fully funded research scholarship in Construction Safety Management.
- Developed Computer Vision models (YOLO) for real-time hazard detection (“Zero Site Accidents”).
- Engineered Digital Twin simulations using Unity to model complex safety scenarios.

Digital Egypt Pioneers (DEPI) Program MCIT Apr 2024 – Oct 2024
AI & Data Science Trainee (IBM Track) Cairo, Egypt

- Mastered advanced Data Science methodologies, including Python, SQL, and Deep Learning.
- Built and deployed a full-stack ML capstone project solving a real-world business problem.
- Applied statistical analysis to extract trends and optimize business decision-making.

Fawry MSME Finance Jul 2024 – Sep 2024
Data Analysis Intern Cairo, Egypt

- Designed executive-level KPI dashboards using Power BI for Marketing and Sales teams.
- Automated reporting workflows using Python and Excel, improving accuracy and efficiency.
- Extracted and preprocessed complex datasets using SQL and Python (Pandas) to ensure high data quality for downstream analysis.
- Collaborated with stakeholders to define key business metrics and translate them into actionable, data-driven insights.

Education

October 6 University 2019 – 2023
B.Sc. Computer Science | GPA: 3.37 / 4.0 | **Military Status: Finally Exempted** Egypt